

QUIZ 05 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday, but are rather designed to be a starting point for your studies.

Use the following setup for Questions 1 through 5:

What is the average temperature across all patients undergoing complex surgery at the Cleveland Clinic? We'll define a complex surgery to be one that lasted for longer than one hour and required general anesthesia. In a 2023 study¹, aspects of several thousands patients undergoing complex surgery at the Cleveland Clinic were recorded. Shown below is the distribution of core body temperatures from a random sample of 120 of these patients, along with some potentially relevant summary statistics.

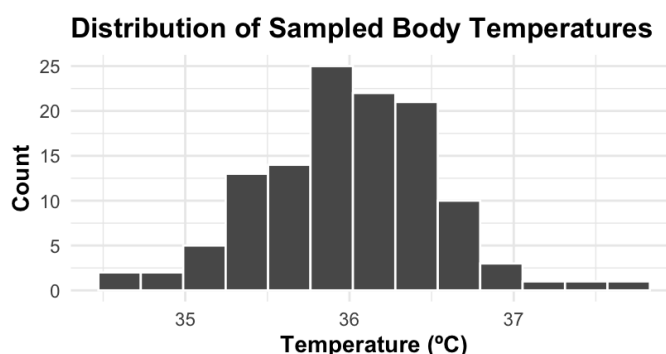


Figure 1: Histogram of Temperatures.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 36.01^\circ\text{C}$	median = 36°C	$s = 0.52^\circ\text{C}$	$n = 120$

Table 1: Summary statistics of the data from the dataset.

1. Which of the following best describes an observational unit in the context of this dataset?
- (A) One person
 - (B) People
 - (C) One complex surgery
 - (D) The population average core body temperature during a complex surgery across all patients at the Cleveland Clinic.

¹Homoki, J. and Nowacki AS, “Core Temperature Dataset”, TSHS Resources Portal (2023)

Solution: One observational unit is one person.

2. Which of the following best describes the parameter of interest, μ , in words?

- (A) An average
- (B) The population average
- (C) The population average temperature
- (D) The population average core body temperature, in Centigrade, during a complex surgery across all patients at the Cleveland Clinic.**

Solution: Remember to not just say “average,” to indicate the population, the actual quantity of interest (core body temperature), and units.

3. Which of the following correctly describes why we are able to proceed with constructing a theory-based confidence interval for μ ?

- (A) Because the sample is random
- (B) Because we have at least 10 successes and at least 10 failures
- (C) Because the sample size is large (at least 20)
- (D) Because the sample size is large (at least 20), and the sample distribution is not too heavily skewed
- (E) Because the sample size of 120 is large (at least 20), and the sample distribution is not too heavily skewed**

Solution: The validity conditions for constructing a theory-based confidence interval for a population mean (which are the same ones we use for conducting a theory-based hypothesis test) are: sample size of at least 20, and a not-too-skewed sample distribution. Choice (D) is not quite correct, though, because, as I’ve mentioned many times, you shouldn’t just say “the sample size is greater than or equal to 20-” you should tell me what the sample size actually *is*, in addition to that!

4. Which of the following correctly computes a theory-based 95% confidence interval for μ ?

- (A) $(36.01) \pm 2 \cdot (0.52)$ (B) $(36.00) \pm 2 \cdot (0.52)$ **(C) $(36.01) \pm 2 \cdot \frac{0.52}{\sqrt{120}}$** (D) $(36.00) \pm 2 \cdot \frac{0.52}{\sqrt{120}}$

Solution: The formula for a 95% confidence interval for μ is

$$\bar{x} \pm 2 \cdot \frac{s}{\sqrt{n}}$$

Plugging in appropriate values yields the correct answer choice.

5. A 90% confidence interval for μ is (35.9313, 36.0887). Consider testing

$$\begin{cases} H_0 : \mu = 36 \\ H_A : \mu \neq 36 \end{cases}$$

using a 0.10 significance level. Would we reject the null hypothesis in favor of the alternative hypothesis, or not?

- (A) Reject the null hypothesis in favor of the alternative hypothesis
- (B) Fail to reject the null hypothesis**
- (C) There is not enough information to determine

Solution: This is essentially a way to test your understanding of the Plausible Values method. That is: our confidence interval provides a range of plausible values for μ . Since 36 - the null value for this problem - is contained in this interval, it is a plausible value for μ meaning we would have failed to reject the stated hypotheses. The fact that we used a 0.10 significance level comes from the fact that the interval provided is associated with a 90% confidence level. I encourage you to take a look at Lecture Activity 11 for more details.

Use the following setup for Questions 6 through 8:

Let's continue the setup of Part 1 above, and continue to explore the sample of 120 patients undergoing complex surgery at the Cleveland Clinic. It turns out that 13 of these patients developed an infection within 30 days of surgery (some were minor, some were slightly more serious). Let's now address the question: what proportion of all patients undergoing complex surgery at the Cleveland Clinic develop an infection within 30 days of their surgery?

6. Which of the following provides the best description of the variable of interest, as it pertains to the original research question?
- (A) The number of patients that developed an infection within 30 days of surgery
 - (B) Whether or not a patient developed an infection within 30 days of surgery**
 - (C) The number of infections a patient developed within 30 days of surgery

Solution: Though choice (C) seems correct, we talked about this during lecture. In particular, this is all part of the "context" of the problem - though I could collect information on the number of infections a patient developed post-surgery, I would then simply compare that number to 1. Stated differently - choice (C) describes a perfectly valid variable that we could collect, choice (B) is the one that "best describes the variable of interest as it pertains to the original question."

7. Are we able to proceed with a theory-based approach to construct a 95% confidence interval for the parameter of interest (which we can call π)?

- (A) Yes, because we took a random sample
- (B) Yes, because the sample size is large
- (C) No, because although our sample size (120) is greater than 20, we do not have the sample distribution and do not know if it is heavily skewed or not.
- (D) Yes, because we have 13 “successes” (patients developing an infection), which is at least 10, and 107 “failures” (patients not developing an infection), which is at least 10.**

Solution: Because the variable is categorical (and our corresponding parameter is a population proportion), we know to check the validity conditions for categorical data which requires at least 10 successes and at least 10 failures.

8. Which of the following is the correct formula to use for constructing a 95% confidence interval for π using the 2SD method?

(A) $\hat{p} \pm 2 \cdot \sqrt{\frac{\pi(1 - \pi)}{n}}$ (B) $\hat{p} \pm 2 \cdot \sqrt{\frac{\pi_0(1 - \pi_0)}{n}}$ (C) $\hat{p} \pm 2 \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$

Solution: This is related to a discussion we had during lecture (and one that was on Lecture Activity 11). The formula in choice (A) cannot be used because it depends on the true value of the parameter π , which is unknown. The formula in choice (B) cannot be used because we don't have a null value π_0 (since we're not conducting hypothesis testing). Choice (C) is correct.

Use the following setup for Questions 9 through 12:

Does walking regularly cause a decrease in cholesterol? Imagine we recruited 20 subjects, and randomly assigned 10 of them to regularly walk every day for a month and assigned the other 10 to maintain relatively sedentary lifestyles for the next month. After 30 days, we measured the cholesterol of each participants - it turns out that the average cholesterol levels of subjects in the “walking regularly” group were quite a bit lower than that of subjects in the “sedentary” group.

9. Is this an experiment or an observational study?

- (A) Experiment** (B) Observational Study

Solution: We explicitly assigned subjects to groups, meaning we conducted an experiment.

10. Can we conclude that the increase in activeness of lifestyle caused the decrease in average cholesterol levels?

- (A) Yes** (B) No

Solution: Because the experiment is randomized, we can establish causal claims (at least, by the conventions we've laid out in the course at this point).

11. 12 subjects among the 20 are classified as “hypertensive” (having high blood pressure). Which of the following is true (select only ONE option):

- (A) We'd expect around 60% of subjects within each treatment group to be hypertensive
- (B) Exactly 60% of subjects within each treatment group will be hypertensive
- (C) Hypertension could have potential confounding effects on our results

Solution: As we saw with the Randomizing Subjects Applet in Lecture Activity 12, randomization doesn't necessarily balance each individual assignment perfectly - as such, choice (B) is not correct. Choice (C) is not correct because we randomized (if we hadn't randomized, then Hypertension could certainly be a potential confounding variable). Choice (A) is the most correct.

Some Potentially Useful Formulas and Definitions

- Confidence Interval = (statistic) $\pm$ (margin of error)
= (statistic) $\pm$ (multiplier) $\cdot$ (standard deviation/error)
- $SD_{\text{NULL}} = \sqrt{\frac{\pi_0(1 - \pi_0)}{n}}$; $SD(\hat{p}) = \sqrt{\frac{\pi(1 - \pi)}{n}}$; $SD(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$
- $SD_{\text{NULL}} = \frac{\sigma}{\sqrt{n}}$; $SD_{\text{NULL}} = \frac{s}{\sqrt{n}}$; $SE(\bar{x}) = \frac{s}{\sqrt{n}}$

You may use the remainder of this page for scratch work; please keep in mind that NOTHING written here will be graded.